Yuxing (Estella) Liu

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Material programming and 4D printing · inverse design with surrogate models · sensor-integrated robotic fabrication

Education

University of Stuttgart, M.Sc. Integrative Technologies and Architectural Design Research (ITECH) — Oct 2025 – present
Selected courses: Distributed Robotics, Computing in Architecture, Agent-Based Modelling, Computational Design and Digital Fabrication — Stuttgart, DE

University College London, The Bartlett, MPhil/PhD in Architectural Design — Oct 2024 – May 2025
Supervisors: Prof. Marjan Colletti, Dr. Kostas Grigoriadis · porosity-driven hygromorphic bilayers — London, UK

University of Pennsylvania, M.S. Design: Advanced Architectural Design (MSD-AAD) — Aug 2022 – Aug 2023
GPA 3.84 / 4.0 · Robotic Fabrication, New Materials, Geometric Structural Design — Philadelphia, US

Suzhou University of Science and Technology, Bachelor of Architecture (5-year) — Sep 2016 – Jun 2022
GPA 3.9 / 4.0, ranked 1st in cohort — Suzhou, CN

University of Hong Kong, Summer School, The SPACE (grade A) · **University of Wisconsin–Madison**, Architecture Summer Program (grade A) — 2019 · 2018

Publications

1. **Liu, Y.**, Huang, Y., & Menges, A. (2026). Gradient Matter: Multi-Objective Inverse Design for 4D-Printed Hygromorphs through Gradient Porosity Programming. *ACADIA 2026*. Peer-reviewed full paper, accepted.
2. Second author (2026). Volumetric Segmentation Strategies for Fused Shell Geometries. *IASS 2026 Symposium*. Full paper, accepted.
3. Second author (2025). Construction-Aware Timber: An Agent-Based Modeling (ABM) Approach for Non-Standard Intersecting Timber Strip Structures. *ACADIA 2025*. Peer-reviewed full paper.
4. **Liu, Y.**, Lu, X., Colletti, M., & Grigoriadis, K. (2025). Controllable Deformation: A 4D-Printing Simulation Framework for Porosity-Driven Hygromorphic Bilayer Structure. *Computational Design and Robotic Fabrication (CDRF 2025)*. Full paper.
5. **Liu, Y.**, Wei, H., Colletti, M., & Grigoriadis, K. (2025). Programming Bilayer Structures in 4D-Printed Hygromorphs by Tailoring Porosity in Macroscale. *CAADRIA 2025*. Published.
6. **Liu, Y.** (2020). The Needs and Strategies for the Adaptation of the Outdoor Environment of Old Relocation Communities. *City House*, 27, 125–128. ISSN 1006-6659.

Research and Work Experience

ICD / IntCDC, Material Programming, Graduate Research Assistant (HiWi) — Feb 2026 – present
ArchiBioFoam project, Horizon Europe EIC Pathfinder — Stuttgart, DE

- Experimental data collection and analysis, 4D-printing calibration and material preparation.

ICD / IntCDC, Large-Scale Construction Robotics Laboratory (LCRL), Graduate Research Assistant (HiWi) — Jan 2026 – present
C# scripting, robotic fabrication, hardware customisation — Stuttgart, DE

ICD / CA, RAW project, Graduate Research Assistant (HiWi) — Nov 2025 – present
Horizon Europe GA no. 101161441 · knowledge graphs and material ontologies, Grasshopper plug-in development — Stuttgart, DE

Rheia Verve, LLC, Interior Designer — Oct 2023 – Oct 2024
3D modelling, rendering and documentation — San Jose, US

Jiangsu Huaxin Urban Planning & Municipal Engineering Design Research Institute, Intern — Jul 2021 – Oct 2021
Preliminary research, 3D modelling and 2D concept drafting — Lianyungang, CN

Teaching

DigitalFUTURES 2026, Teaching Assistant — *ReForm: Exploring 4D Printing in Architecture* — Jun 2026
In-person workshop: teaching support and student guidance through final reviews

The Bartlett, UCL, Guest Lecturer — *Houdini Workflow of Half-Symmetry Generation* (with Richard Beckett) — Jan 2024
Two-day online workshop: instruction, assignments and final critique

Workshops

ACADIA 2026, Soft-Hard Programmable Furniture — *digital creation; multi-material 3D printing, Ovenbird toolpaths* — 2026

DigitalFUTURES 2025, Deployable Structures with Vector-based Graphic Statics — *physical models, digital drawings* — 2025

CAADRIA 2025, Kit-of-Parts Gridshell — *physical models; ECNC networks, timber construction* — 2025

ACADIA 2024, Digital Deformation: Design-to-Fabrication in Ceramic 3D Printing — *G-code and toolpath design (WASP)* — 2024

DigitalFUTURES 2024, Prompt-to-Production: Automation for Circular Timber Assemblies — *algorithm design, robotic fabrication* — 2024

Honours and Awards

Top Graduate Award (rank 1 in cohort), Suzhou University of Science and Technology — 2022

Second Prize, 4th Architectural Creation Competition, *Recreational Life 2.0* (team leader) — 2021

National Scholarship for Academic Excellence, Ministry of Education, China (3×) — 2018 – 2020

First host, Jiangsu Students' Innovation and Entrepreneurship Training Program, *Livable Community for the Elderly* — 2019 – 2020

Second Prize, National Four-School Architecture Joint Design, *Remembrance* (team leader) — 2019

Skills

Programming	Python (NumPy, SciPy, scikit-learn, PyTorch, pandas), C# (RhinoCommon / Grasshopper plug-ins), VEX
Robotics	KUKA (KRL, KUKA|prc), Universal Robots (URScript), Arduino firmware and sensors, TCP calibration
Fabrication	Multi-material FDM, custom G-code / Ovenbird / CuraEngine toolpaths, WASP clay printing, binder jetting, robotic hot-wire cutting, CNC milling, laser cutting
Design and simulation	Rhino, Grasshopper, Houdini, Karamba3D, Kangaroo, Blender, Revit
Languages	Chinese (native), English (academic; degrees in the US and UK), German (A1–A2)

Updated November 2026